

Arrays

Arrays are collection of same data typed elements in a single variable.

In array, memory allocation will be sequential. i.e all the elements will store side by side.

Variable Declaration:

```
data_type variable_name;
```

```
int i;
```

```
float area;
```

Array variable declaration:

Syntax: - datatype variable_name[];

```
datatype[] variable_name = new datatype[size];
```

Size= number of elements. And size must be an positive integer.

```
int sub[5]; //student having five subjects.
```

This is also called single dimension array.

```
int a[5]; // integer array
```

```
float avg[10]; // float array
```

```
double sal[100]; // double array
```

```
char name[20]; // Character array. (String)
```

Initialization of array.

```
Int a[]={56,89,3,654,85};
```

Saving of array.

Name=a;

	56	89	3	654	85
Index	a[0]	a[1]	[2]	[3]	[4]
int	1000	1004	1008	1012	1016
float	1000	1004	1008	1012	1016
double	1000	1008	1016	1024	1032

0,1,2,3,4 are called index.

in arrays index always starts from 0.

a[0]=>56;

a[1]=>89;

Print(a[0]); -> 56

Print(a[2]); ->3

Print(a[5]); -> Error

Printing of array elements



W.A.P to declare, initialize and printing of an array.

W.A.P to find the sum of all the array elements.

W.A.P to print all even numbers of an array

W.A.P to find maximum and minimum element of array

W.A.P to search an element in an array.

W.A.P to find the occurrence of given element of an array.

W.A.P to find the occurrence of each element of a given array.

W.A.P. to delete an element in an array.

W.A.P. to insert an element in an array.

W.A.P to sort an array in ascending order.

2	3	1	4	1
Index a[0]	a[1]	2	3	4

I	SUM	S=5
0	0+2 -> 2	
1	2+3->5	
2	5+1->6	
3	6+4->10	
4	10+1->11	
5		

Merging/addition of two array;

A	B	c
5	6	11
4	2	6
9	5	14
6	6	12
3	6	9
6		

5	6	8	2	6	10
0	1	2	3	4	5

max=a[0]; -> max=5

if(a[i]>max)

max=a[i]

max=a[0]; 5, 6,8,9,,10

min=a[0];5,2

5 > 6

5	6	8	9	6	10	2
0	1	2	3	4	5	6
2	6	8	9	6	10	5
2	5	8	9	6	10	6

2	5	6	9	8	10	6
2	5	6	6	9	10	8
2	5	6	6	8	10	9
2	5	6	6	8	9	10

if(a[3]>a[2])

{

swap;

}

ele = 20

po =3

5	6	8	2	6	10
0	1	2	3	4	5
2	6	8	5	6	10
2	5	8	6	6	10
2	5	6	8	6	10
2	5	6	6	8	10

/*W.A.P. to delete an element in an array*/

5	6	8	10	6	20	9
0	1	2	3	4	5	6

Ele = 10

Po = 3

For(j=po;j<s;j++)

$a[j] = a[j+1]$

$s--;$

5	6	2	6	10	10
0	1	2	3	4	5

/*W.A.P. to insert an element in an array*/

5	6	8	2	6	10
0	1	2	3	4	5

Ele = 20

Po = 3

5	6	20	8	2	6	10
0	1	2	3	4	5	6

$A[po]=ele;$

5	-6	8	-20	2	-4	10
0	1	2	3	4	5	6

Two Dimensional Array.

Declaration: type name[rows][columns];

```
int a[3][3]={3,4,5},{5,6,8},{4,9,5}};
```

total number of elements = rows X columns = 3 X 3 = 9

a	0	1	2
0	3 a[0][0]	0 a[0][1]	0 a[0][2]
1	5 a[1][0]	6 a[1][1]	0 a[1][2]
2	4 a[2][0]	9 a[2][1]	5 a[2][2]

```
for(i=0;i<r;i++)
```

```
{
```

```
for(j=0;j<c;j++)
```

```
printf("%d ",a[i][j]);
```

```
} l = 6 , 6+1, 6+2,6+3
```

h	e	r	a	g		r	a	g	h	u		s	k
0	1	2	3	4	5	6	7	8	9	10	11	12	13

J=0,1,2,3

r	a	g	h	u	'\0'				
0	1	2	3	4	5				

I		a	m		a		S	t	u	d	e	n	t	'\0'			
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14			